Activation Energy and Catalysts

Energy profiles, temperature, the Arrhenius equation and catalysts

Name _____ Class _____ Date _____

USEFUL RELATIONSHIPS

$$k = A\,e^{-E_a/RT} \qquad \ln\frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \qquad R = 8.314\ \text{J/(mol·K)}$$

· 1 kJ = 1000 J · T in kelvin

Convert Ea to J/mol before substituting, and show units in every step.

A. Understanding

1. Define activation energy and transition state.

2. A reaction has Ea (forward) = 120 kJ/mol and ΔH = −40 kJ/mol. (a) Is it exothermic or endothermic? (b) Find Ea for the reverse reaction. (c) A catalyst lowers the forward Ea to 70 kJ/mol. What is the reverse Ea now, and what is ΔH?

3. Use the Maxwell–Boltzmann distribution to explain why a small increase in temperature can roughly double the rate of a reaction.

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4. Explain the difference between a homogeneous and a heterogeneous catalyst, with one example of each, and state the effect of a catalyst on K.

B. The Arrhenius equation

5. A reaction has Ea = 75.0 kJ/mol. By what factor does k increase between 300. K and 310. K?

6. A rate constant is $3.0 \times 10^{-5}\ \text{s}^{-1}$ at 500. K and $6.0 \times 10^{-4}\ \text{s}^{-1}$ at 550. K. Calculate Ea in kJ/mol.

7. For a reaction with Ea = 75.0 kJ/mol, $k = 2.0 \times 10^{-4}\ \text{s}^{-1}$ at 300. K. Calculate k at 330. K.

C. Challenge

8. A catalyst lowers Ea from 75.0 kJ/mol to 50.0 kJ/mol at 298 K. Assuming the frequency factor A is unchanged, by what factor does the rate constant increase?

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9. An Arrhenius plot of ln k against 1/T is a straight line with slope -6.65×10^3 K. Calculate Ea.

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